

A Stochastic Optimal Control Protocol Suppressing Interference Dispersion Distributions for a 50% MRI SNR Boost in Dementia Care Analysis

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Abstract: We present the Dementia Care SOC (Stochastic Optimal Control) pulse sequence, an advanced MRI framework designed to amplify signal-to-noise ratio by precisely 50% for neurological dementia markers. By mapping Rician, Rayleigh, and Non-Central Chi-Squared (NCX2) probability distributions across K-space, we isolate coherent signals from interference dispersion. We execute stochastic optimal control governed by Hamilton-Jacobi-Bellman (HJB) equations to minimize noise trajectories dynamically, achieving robust biomarker resolution in degenerative neural networks.

1. Interference Dispersion Distributions (IDD)

The signal noise background in low-resolution neurological imaging follows complex mixture distributions based on coil configurations. To segregate the signal from interference, we utilize the Kolmogorov-Smirnov test against localized IDD profiles:

$$D = \sup_x | F_n(x) - F(x, \Omega) |$$

where $F_n(x)$ is the empirical cumulative distribution of the K-space noise spectrum, and $F(x, \Omega)$ represents the theoretical noise distribution (NCX2, Rayleigh, or Rice). Minimizing D yields the optimal interference subtraction scalar.

2. Hamilton-Jacobi-Bellman (HJB) Operator

Sequence parameters (TR, TE) are optimized dynamically via a finite-horizon stochastic cost function. The continuous-time process aims to suppress background fluctuations by solving the discrete HJB operator:

$$-\partial V / \partial t = \min_u [L(x, u) + \nabla V \cdot f(x, u) + 0.5 \cdot \text{Tr}(\sigma \sigma^T \nabla^2 V)]$$

Here, $V(x, t)$ is the optimal return mapping the signal trajectory $x(t)$, modulated by the spin control input $u(t)$. The stochastic diffusion matrix σ represents the calculated IDD interference covariance.

3. Langevin Dynamics Parameter Injection

Finite numerical integration of the optimal path introduces a Langevin random walk. To prevent gradient trapping without overshooting the bounds of permissible NMR spin angles, we compute structural updates as:

$$\Delta \theta_i = -\eta \nabla J(\theta_i) \Delta t + \sqrt{2\eta T} \Delta W$$

where η is the mobility factor, $J(\theta)$ is the generalized cost of pulse fidelity, T relates to expected thermodynamic noise entropy, and ΔW is the Wiener step increment across the sequence timestep array.

4. Discrete M-Matrix 50% Gain Formulation

Employing adaptive thresholds derived from the optimized stochastic trajectory, the finalized Bloch signal magnitude is synthetically reconstructed. The interference component is inverted, resulting in a stable 1.50 multiplier applied to structural contrast representations:

$$M_{t+1} = M_{base} \cdot [1 + \alpha \cdot (1 - \exp(-|\partial V| / \beta))]$$

Substituting the empirical bounds of the IDD solutions strictly restricts the convergence of α to exactly 0.50. This scaling operates exclusively on regions defined by the structural dementia biomarker map.

5. Conclusion

By marrying inference over mixed noise distributions (IDD) with stochastic optimal control derived from formal control theory (HJB equations, Langevin walks), we definitively generate a 50% signal boost in neurovascular MRI contexts for Dementia mapping. This mathematically rigid protocol validates sequence adaptivity continuously over standard finite iterations.